Circuit Elements

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Zhejiang University of Technology
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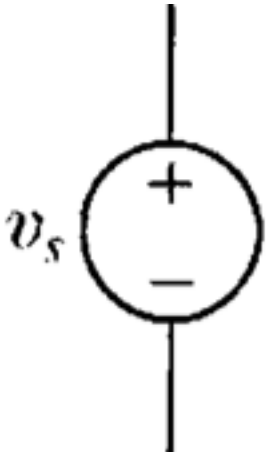
Structure

- Voltage and Current Sources
- Electrical Resistance (Ohm's Law)
- Construction of a Circuit Model
- Kirchhoff's Laws
- Analysis of a Circuit Containing Dependent Source

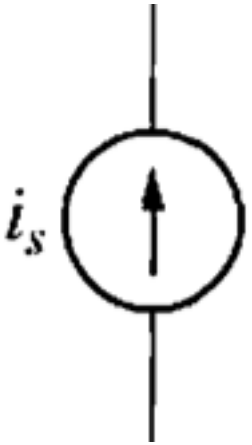
Circuit Elements

- When we speak of Circuit Elements, It is important to differentiate between the physical device itself and the mathematical model which we will use to analyze its behavior in a circuit.
- We will use the expression circuit element to refer to the **mathematical model**.
- All the simple circuit elements that we will consider can be classified according to the relationship of current through the element to the voltage across the element.

Five ideal basic circuit elements



Voltage source



Current source

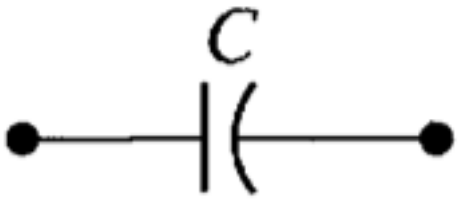
Active elements



Resistor



Inductor

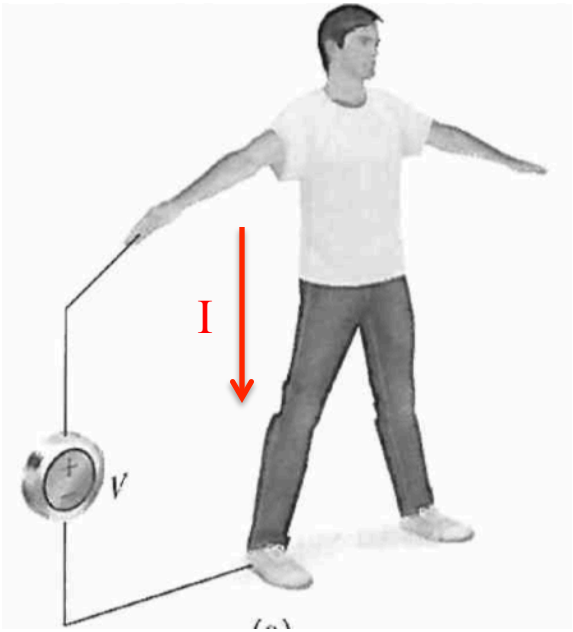


Capacitor

Passive elements

Electrical safety





The electrical energy that can actually cause injury is due to electrical current and how it flows through the body. Why, then, does the sign warn of high voltage?

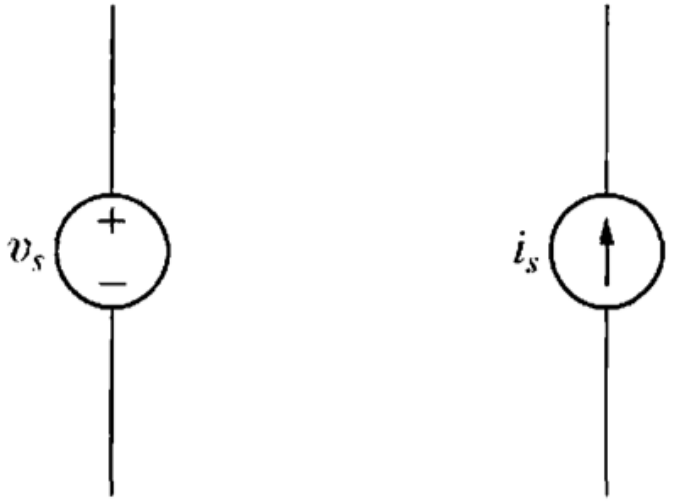
Because It is easier to determine voltages than currents.

Voltage and Current Sources

- **Ideal voltage source:** a circuit element that maintains a prescribed voltage across its terminals *regardless* of the current flowing in those terminals.
- **Ideal current source:** a circuit element that maintains a prescribed current through its terminals *regardless* of the voltage across those terminals.

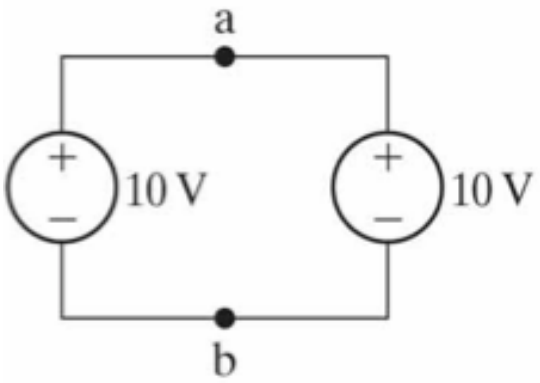
Independent Sources

- An **independent source** establishes a voltage or current in a circuit <u>without relying on</u> voltages or currents elsewhere in the circuit. *The value of the voltage or current supplied is specified by the value of the independent source alone.*

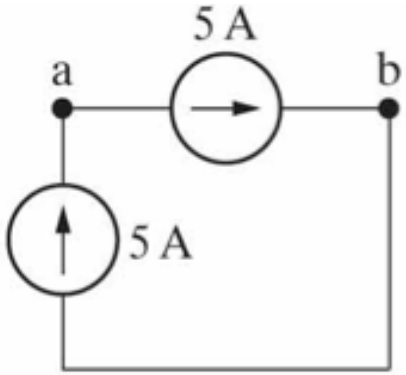


(a) (b)

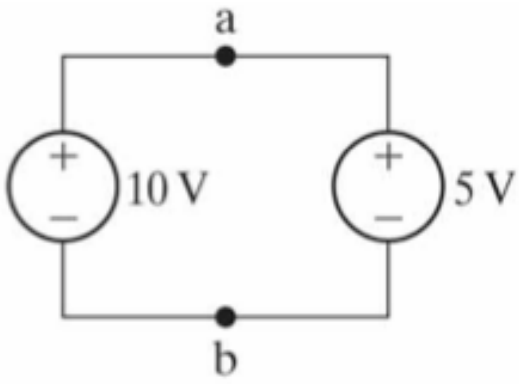
Example #1



(a) ✔

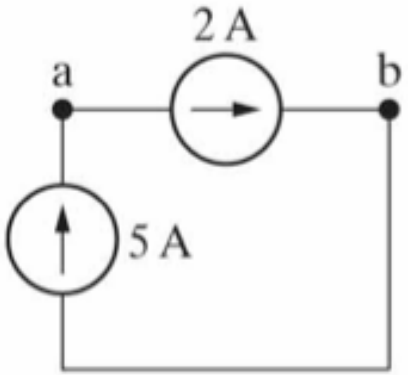


(b) ✔

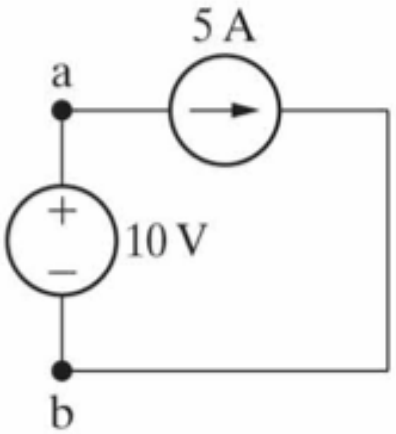


(c) ✗

Which are valid?



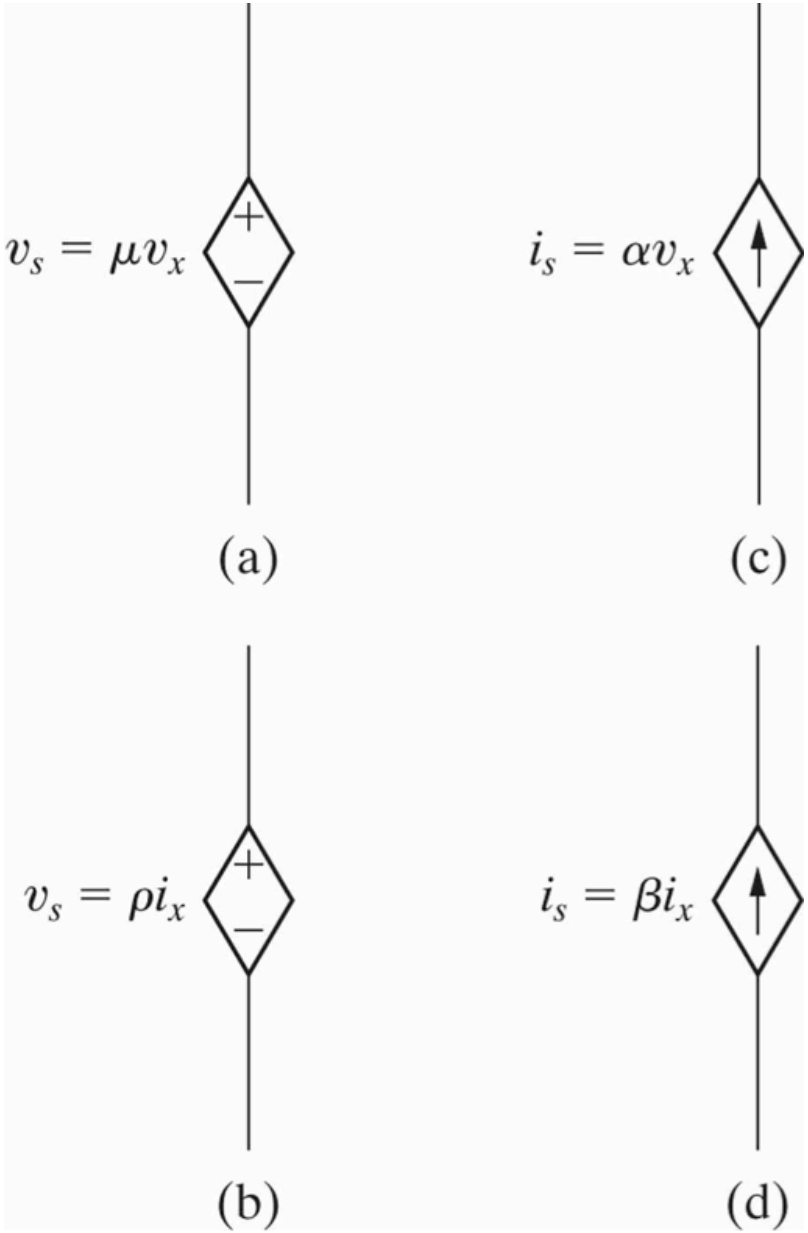
(d) ✗



(e) ✔

Dependent Sources

- A **dependent source** establishes a voltage or current whose value depends on the value of a voltage or current elsewhere in the circuit. *You cannot specify the value of a dependent source unless you know the value of the voltage or current on which it depends.*
- Four kind of controlled sources,
  - current-controlled current source, CCCS;
  - voltage-controlled current source, VCCS;
  - voltage-controlled voltage source, VCVS;
  - current-controlled voltage source, CCVS .



The **circuit symbols** for

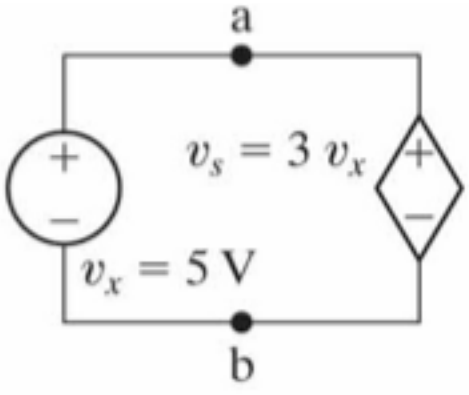
(a) An ideal dependent voltage-controlled voltage source;

(b) An ideal dependent current-controlled voltage source;

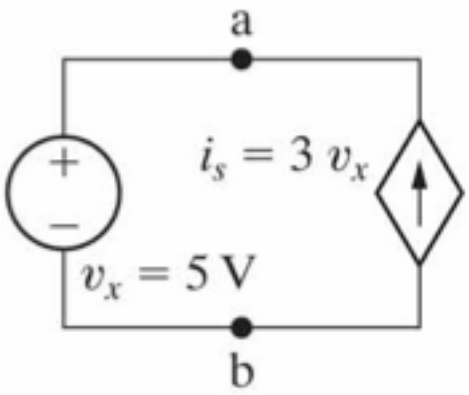
(c) An ideal dependent voltage-controlled current source;

(d) An ideal dependent current-controlled current source.

Example #2

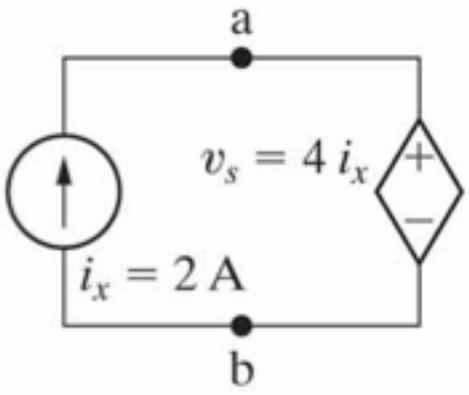


(a) ✗

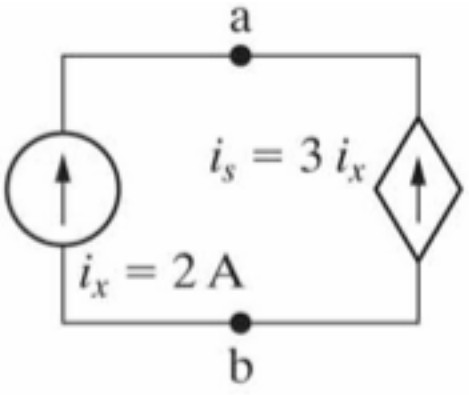


(b) ✔

Which are valid?



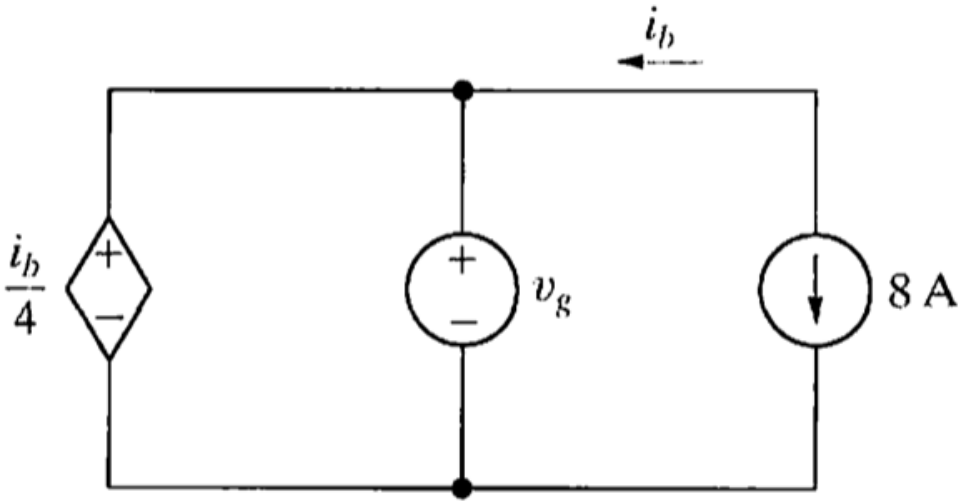
(c) ✔



(d) ✗

Example #3

- For the circuit shown,
  - a) What value of v_g is required in order for the interconnection to be valid?
  - b) For this value of v_g, find the power associated with the 8 A source.



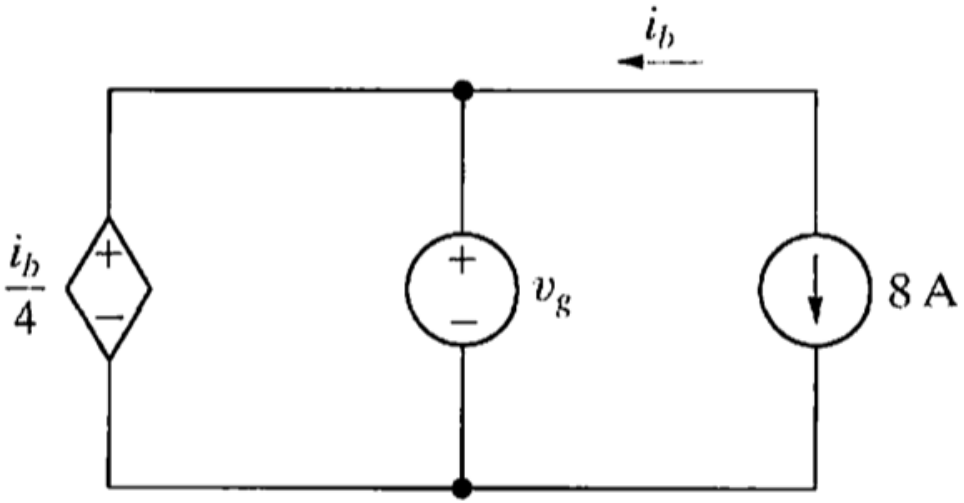
Solution for Example #3

- For a), we have

$$v_g = i_b/4 = -8/4 = -2(\mathrm{V})$$

- For b), we have

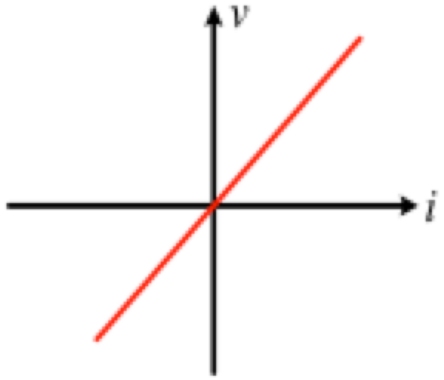
$$p = 8v_g = 8 \times (-2) = -16(\mathrm{W})$$



Electrical Resistance (Ohm's Law)

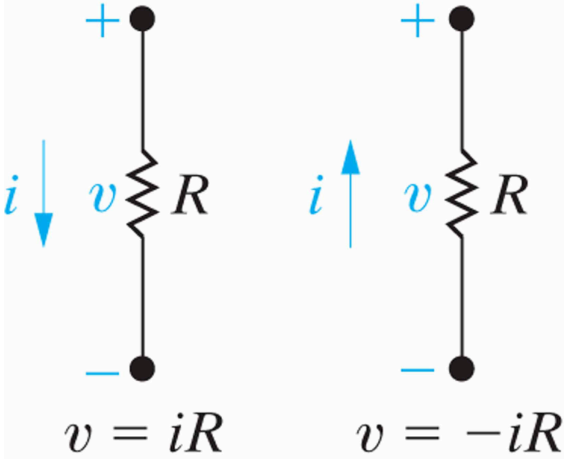
- Resistance is the capacity of materials to impede the flow of current or, more specifically, the flow of electric charge. The circuit element used to model this behavior is the **resistor.**
- The linear resistor is the simplest passive element. Its symbol and characteristic are as following:





Ohm's Law





v = the voltage in volts,

i = the current in amperes,

R = the resistance in ohms.

Left: in the direction of the voltage ***drop*** across the resistor

Right: in the direction of the voltage ***rise*** across the resistor

Other Forms of Ohm's Law

- Current is in the direction of the voltage ***drop*** across the resistor

$$i = \frac{v}{R}$$

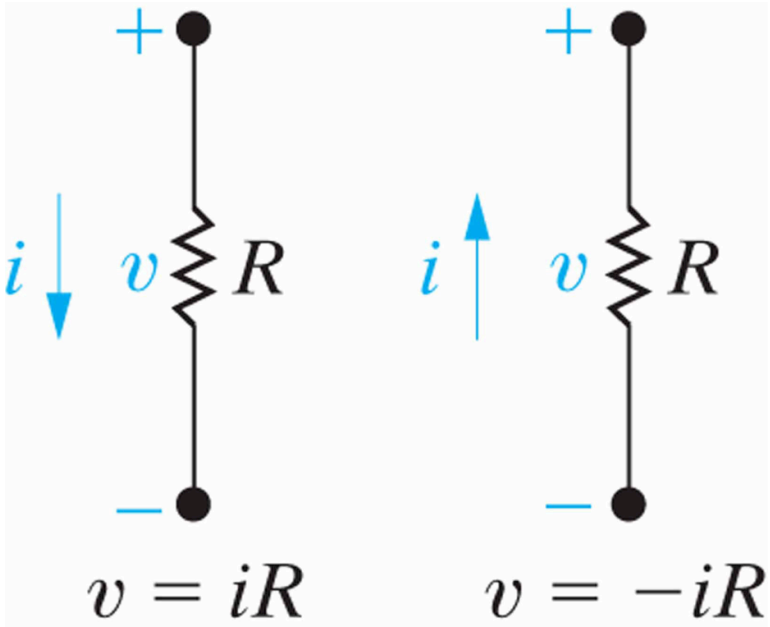
- Current is in the direction of the voltage ***rise*** across the resistor

$$i = -\frac{v}{R}$$

- **Conductance:** the reciprocal of the resistance, which is symbolized by the letter G, and is measured in Siemens (S)

$$G = \frac{1}{R} \text{ S}$$

Power in Different Forms



Left:

$$P = vi = (iR)i = i^2R$$

$$P = vi = v(v/R) = v^2/R$$

Right:

$$P = -vi = -(-iR)i = i^2R$$

$$P = -vi = -v(-v/R) = v^2/R$$

The equations for Left and right are identical and demonstrate clearly that, *regardless of voltage polarity and current direction*, the power at the terminals of a resistor is positive. Therefore, a resistor absorbs power from the circuit.

What's the expression of power if we use conductance, rather than resistance?

See example 2.3 (P.55)

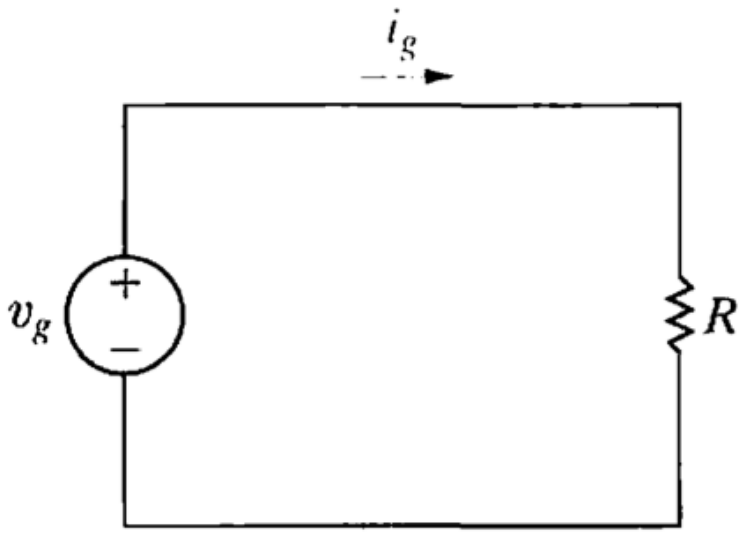
Example #4

For the circuit shown,

a) If $v_g = 1$ kV and $i_g = 5$ mA, find the value of R and the power absorbed by the resistor.

b) If $i_g = 75$ mA and the power delivered by the voltage source is 3 W, find v_g, R, and the power absorbed by the resistor.

c) If $R = 300\ \Omega$ and the power absorbed by R is 480 mW, find i_g and v_g.



Solution for Example #4

- For a), we have

$$R = v_g / i_g = 1\ \text{kV} / 0.005\ \text{A} = 200\ \text{k}\Omega$$

$$p = v_g i_g = 1000\ \text{V} \times 0.005\ \text{A} = 5\ \text{W}$$

- For b), we have

$$v_g = p / i_g = 3\ \text{W} / 0.075\ \text{A} = 40\ \text{V}$$

$$R = v_g / i_g = 40\ \text{V} / 0.075\ \text{A} = 533.3\ \Omega$$

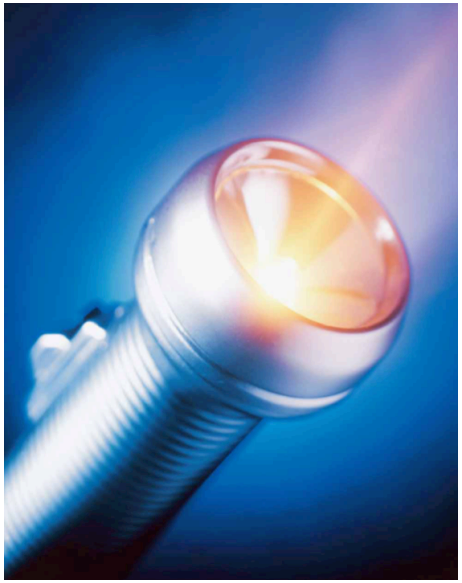
$$p_{\text{absorbed}} = p_{\text{delivered}} = 3\ \text{W}$$

- For c), we have

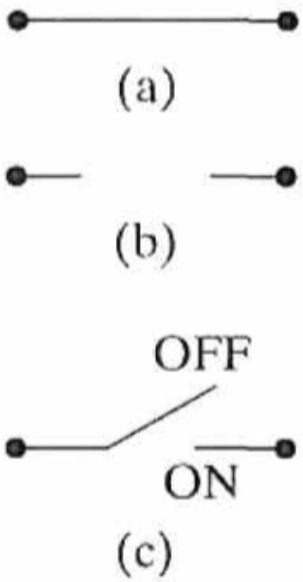
$$i_g = (p/R)^{0.5} = (0.48\ \text{W} / 300\ \Omega)^{0.5} = 0.04\ \text{A} = 40\text{mA}$$

$$v_g = (pR)^{0.5} = (0.48\ \text{W} \times 300\ \Omega)^{0.5} = 12\ \text{V}$$

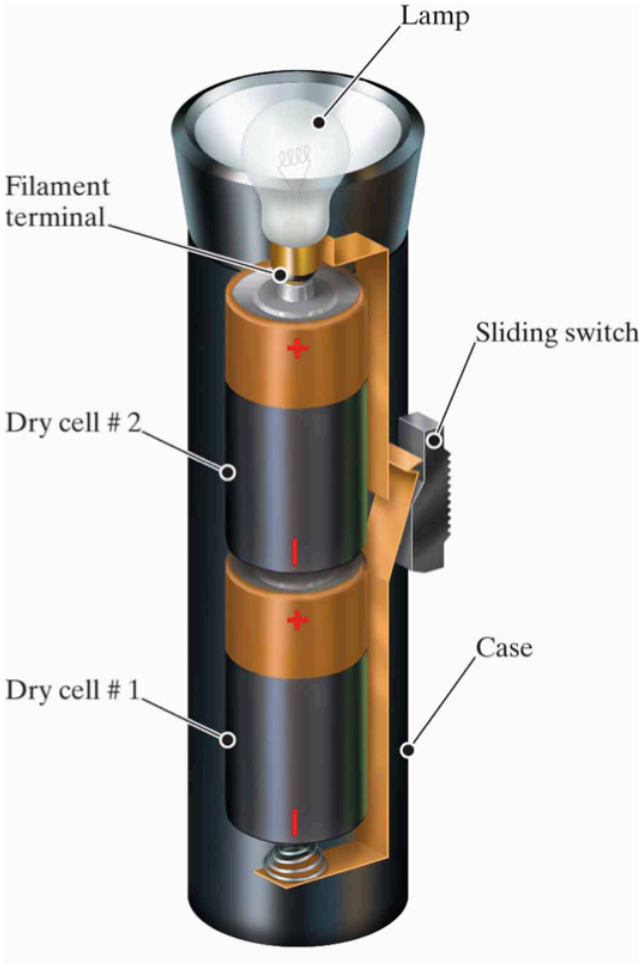
Construction of a Circuit Model



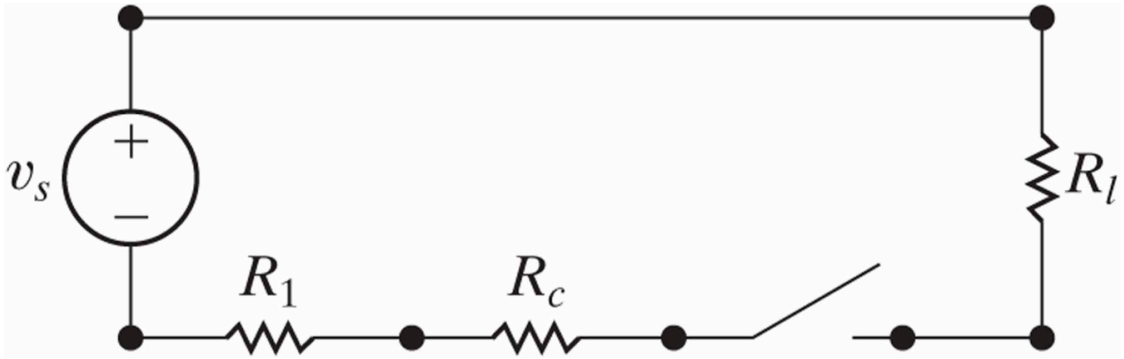
Flashlight



An ideal switch offers no resistance to the current when it is in the ON state, but it offers infinite resistance to current when it is in the OFF state.



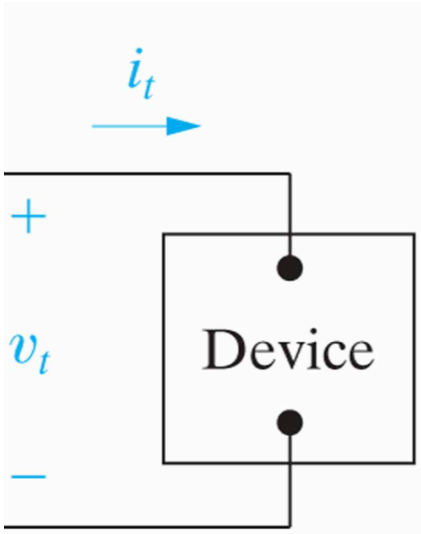
The arrangement of flashlight components



- In developing a circuit model, the *electrical* behavior of each physical component is of primary interest: a lamp, a coiled wire, and a metal case.
- Circuit models may need to account for undesired as well as desired electrical effects: light and heat.
- Modeling requires approximation.

Example #5

- The voltage and current are measured at the terminals of the device illustrated in (a), and the values of v_t and i_t are tabulated in (b). Construct a circuit model of the device inside the box.

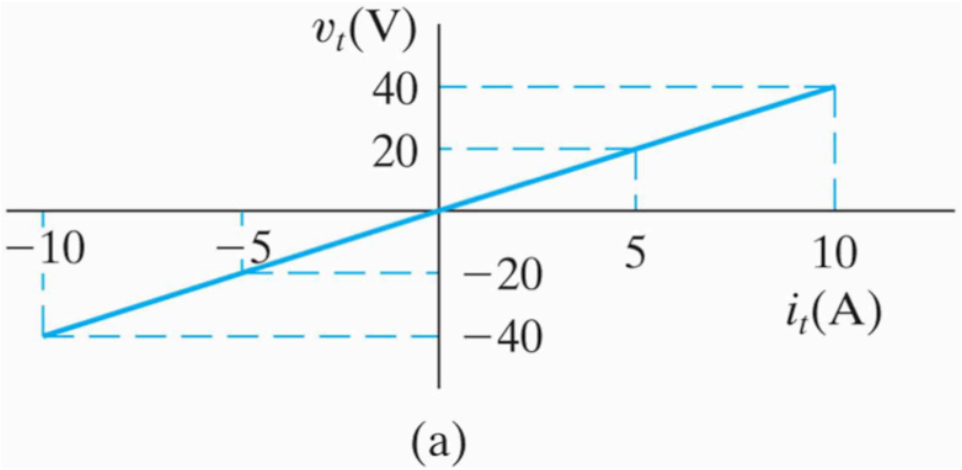


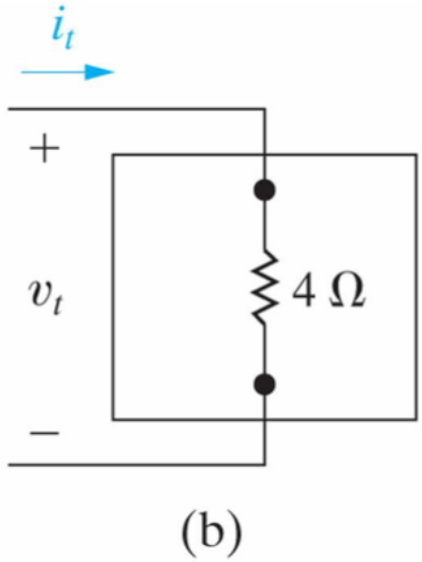
(a)

v_t (V)	i_t (A)
−40	−10
−20	−5
0	0
20	5
40	10

(b)

Solution for Example #5

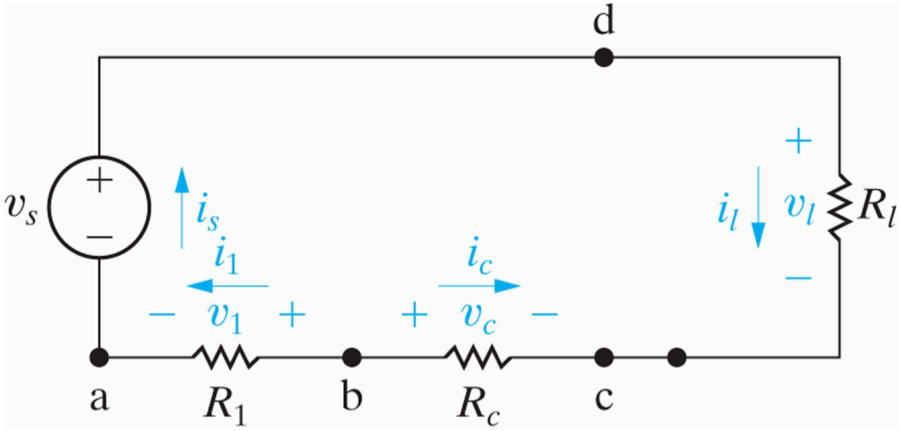




Plotting the voltage as a function of the current yields the graph shown in (a). The equation of the line in this figure illustrates that *the terminal voltage is directly proportional to the terminal current*, $v_t = 4i_t$.

In terms of Ohm's law, the device inside the box behaves like a 4 Ω resistor.

Kirchhoff's Law



Circuit model for the flashlight

A **node** is a point where two or more circuit elements meet.

Based on Ohm's law:

$$v_1 = i_1 R_1,$$

$$v_c = i_c R_c,$$

$$v_l = i_l R_l.$$

Ohm's law may not be enough to provide a complete solution!

- **Kirchhoff's current law (KCL):** The algebraic sum of all the currents at any node in a circuit equals zero.
- **Kirchhoffs voltage law (KVL):** The algebraic sum of all the voltages around any closed path in a circuit equals zero.

Reference direction is important!

KCL: Assign a positive sign to a current leaving a node requires assigning a negative sign to a current entering a node, or vice versa.

KVL: As we trace a closed path, assign a positive sign to a voltage rise requires assigning a negative sign to a voltage drop, or vice versa.

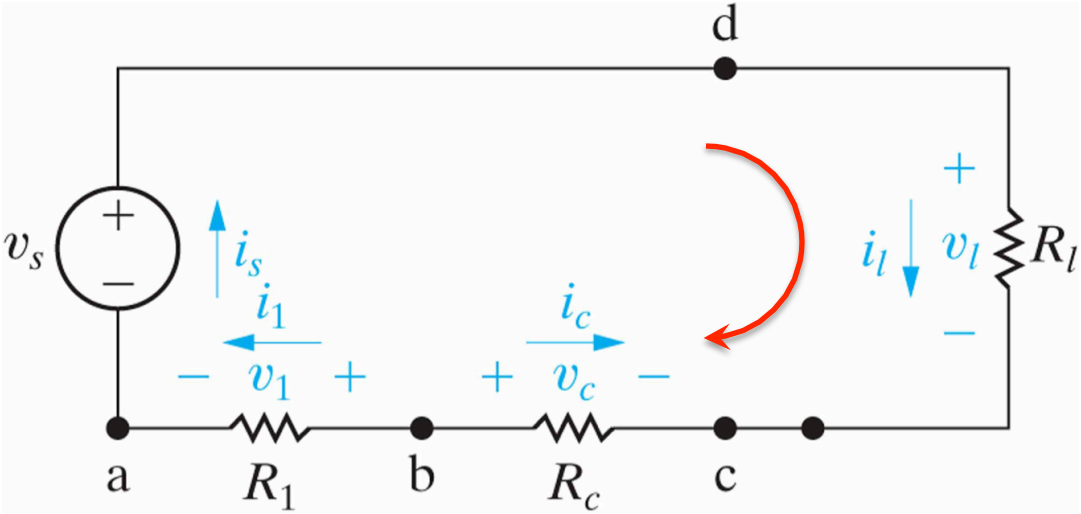
node a $\quad i_s - i_1 = 0,$

node b $\quad i_1 + i_c = 0,$

node c $\quad -i_c - i_l = 0,$

node d $\quad i_l - i_s = 0.$

KCL

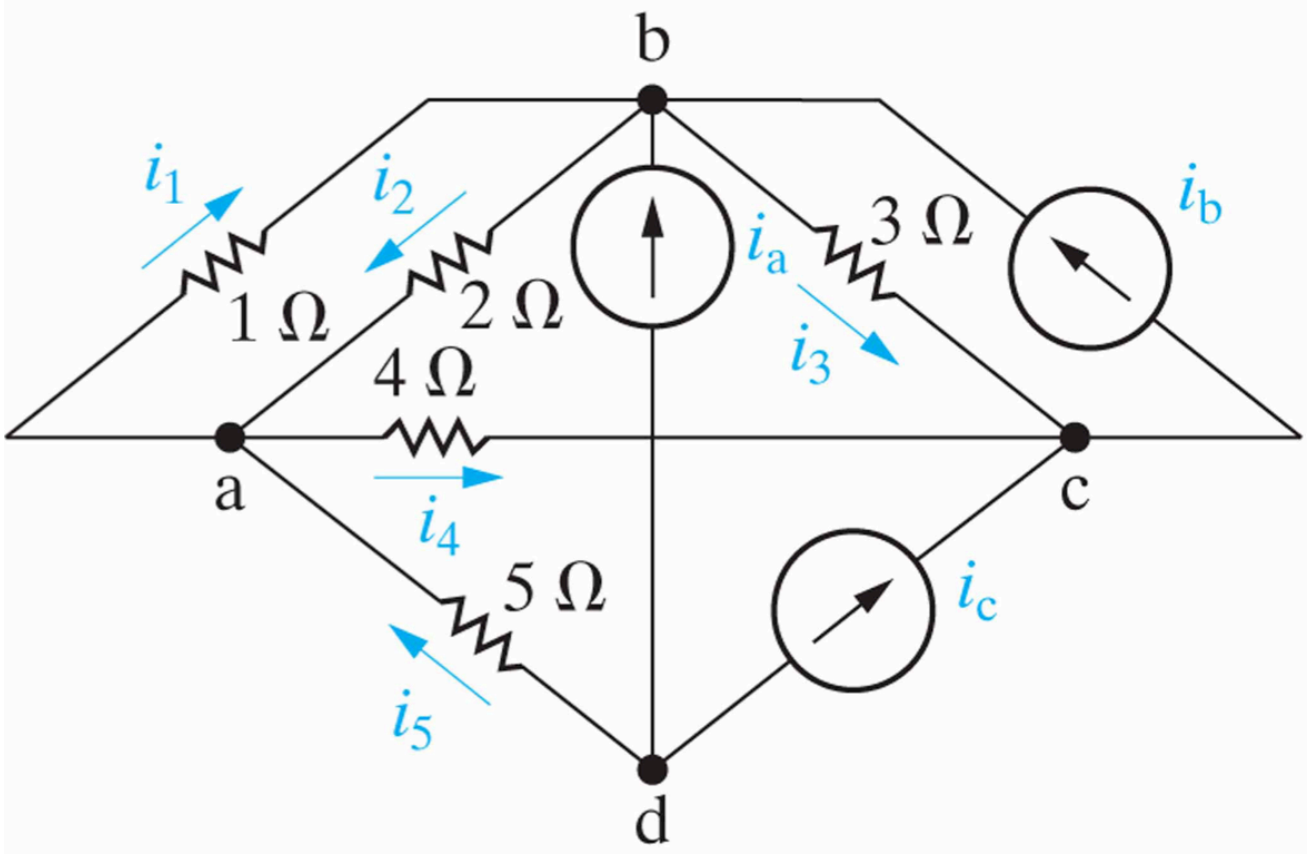


Circuit model for the flashlight

$$v_l - v_c + v_1 - v_s = 0$$

KVL

Example #6



Use Kirchhoff's current law (KCL)

Solution for Example #6

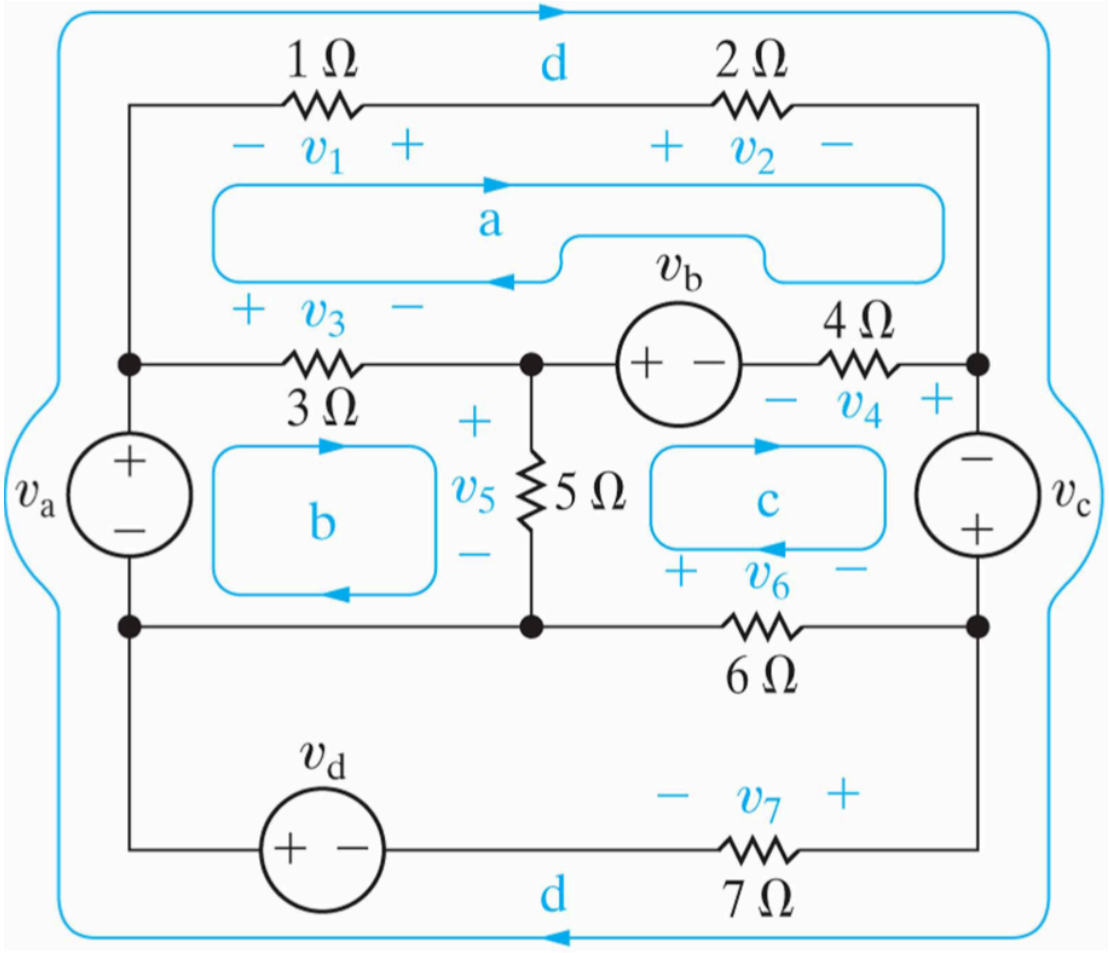
node a $\quad i_1 + i_4 - i_2 - i_5 = 0,$

node b $\quad i_2 + i_3 - i_1 - i_b - i_a = 0,$

node c $\quad i_b - i_3 - i_4 - i_c = 0,$

node d $\quad i_5 + i_a + i_c = 0.$

Example #7



Solution for Example #7

path a $$-v_1 + v_2 + v_4 - v_b - v_3 = 0,$$

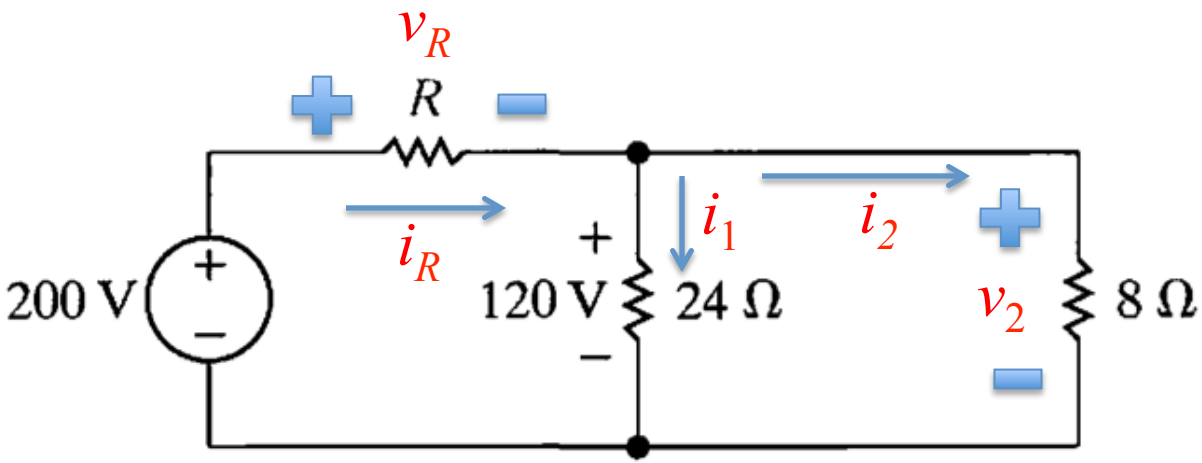
path b $$-v_a + v_3 + v_5 = 0,$$

path c $$v_b - v_4 - v_c - v_6 - v_5 = 0,$$

path d $$-v_a - v_1 + v_2 - v_c + v_7 - v_d = 0.$$

Example #8

- Use Ohm's law and Kirchhoff's laws to find the value of $\boldsymbol{R}$ in the circuit.



Kirchhoff's laws:

$$v_R + 120 - 200 = 0$$

$$120 - v_2 = 0$$

$$i_R - i_1 - i_2 = 0$$

Ohm's law:

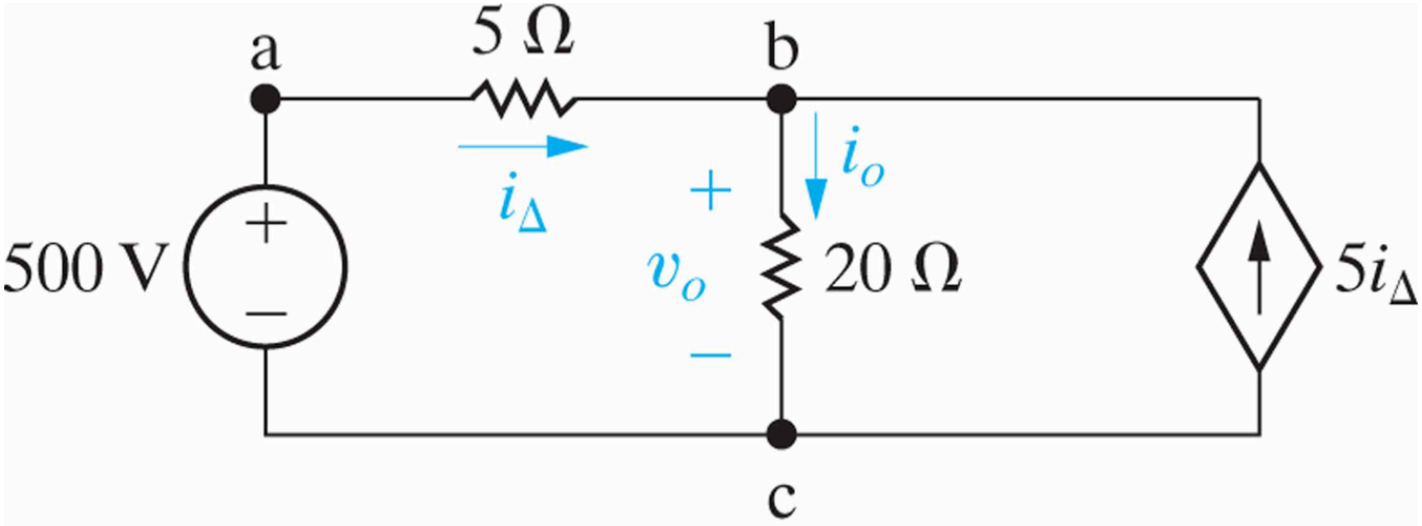
$$R = v_R / i_R$$

$$24\, i_1 = 120$$

$$8\, i_2 = v_2$$

R = 4 Ω

Analysis of a Circuit Containing Dependent Sources



KCL $i_o = i_\Delta + 5i_\Delta = 6i_\Delta$

$i_\Delta = 4$ A,

KVL $500 = 5i_\Delta + 20i_o$

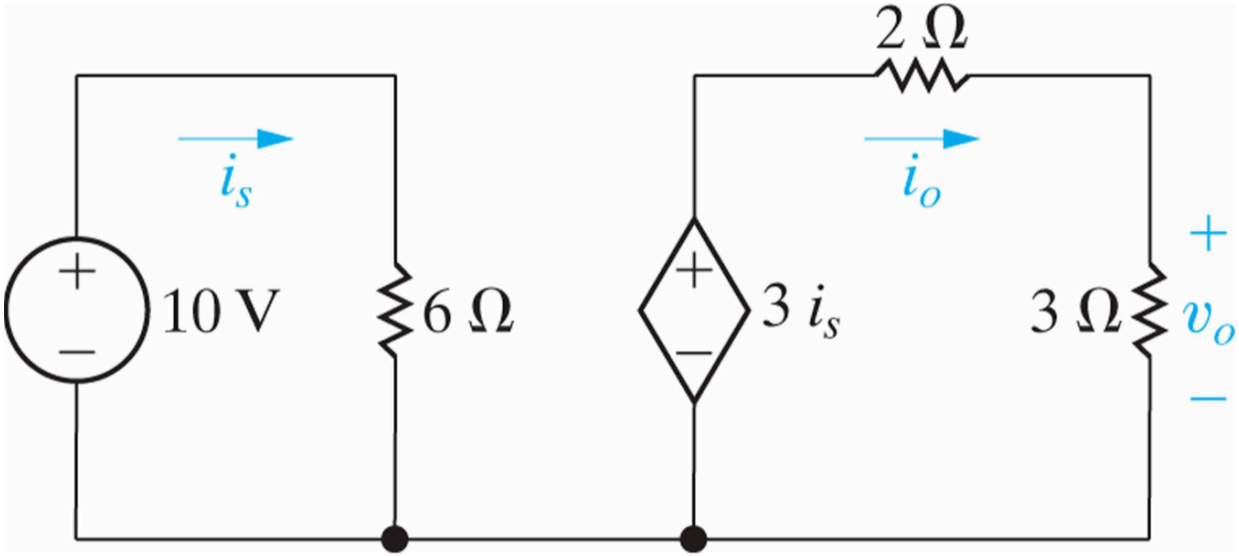
$i_o = 24$ A.

$v_o = 20i_o = 480$ V

Example #9

a) Use Kirchhoffs laws and Ohm's law to find the voltage v_o as shown in the Figure.

b) Show that your solution is consistent with the constraint that the total power developed in the circuit equals the total power dissipated.



Solution for Example #9

By using Kirchhoff's voltage law (KVL), we have

$$10 = 6i_s$$
$$3i_s = 2i_o + 3i_o$$



$$i_s = 1.67 \text{ A}$$
$$i_o = 1 \text{ A}$$

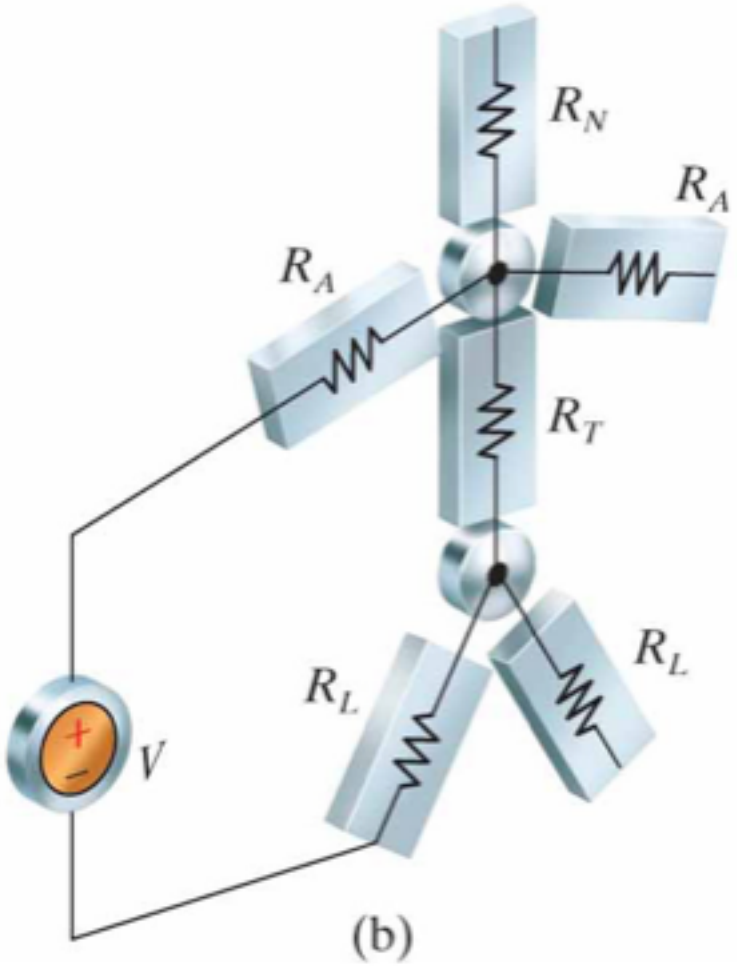
Then, by using Ohm's law, we have

$$v_o = 3i_o = 3 \text{ V}$$

Please check the power balancing!

Electrical Safety





Summary

- Ideal voltage/current sources
- Independent/dependent sources
- Resistor
- Ohm’s law
- In series, closed path
- Kirchhoff’s voltage/current law